Literal chiral markings in $\mathrm{PSU}_4(2)$

Human Compression project

We prove a positive lower bound using explicit permutations and relation words. The finite search which found them is not a premise of this lower bound. The named group is the acquired degree-forty model of $\mathrm{PSp}_4(3) \cong \mathrm{PSU}_4(2)$, of order 25920; its identification is retained from the reviewed subgroup atlas. In particular, it is not identified by its order alone.

We use the already acquired geometric recognition theorem of Schulte and Weiss, in the prefix coordinates of our reviewed manuscript. At rank three the tuple $(1, s, t)$ represents a chiral marking when

$$\langle s, t\rangle = G, \quad |t| = 2, \quad |s|, |s^{-1}t| > 2,$$

the three cyclic subgroups $\langle s\rangle$, $\langle t\rangle$, $\langle s^{-1}t\rangle$ intersect pairwise trivially, and no automorphism simultaneously inverts s and t. The cyclic subgroups are precisely the three nontrivial proper difference subgroups. The remaining difference subgroups are 1 and G, so these tests imply the full intersection property. We count ordered markings modulo $\mathrm{Aut}(G)$, retaining the ordered type and distinguishing mirror markings. This convention is explicit throughout.

Lemma 1. *Suppose s, t and s', t' each generate the same finite permutation group G. Start with the proposed image $1 \mapsto 1$, and propagate images by the directed edges $x \mapsto xs$ and $x \mapsto xt$, using target edges $y \mapsto ys'$ and $y \mapsto yt'$. If propagation is consistent on every edge, it defines the unique automorphism sending (s, t) to (s', t'). If it is inconsistent, there is an explicit word w with $w(s, t) = 1$ and $w(s', t') \neq 1$.*

Proof. Every element is a positive word in s, t, since inverses are positive powers in a finite group. The directed graph is therefore connected from the identity. Consistency means that the proposed image is independent of the chosen positive word: follow the letters of any such word along the verified edges. Consequently concatenating words shows that the resulting function preserves multiplication. It is surjective because s', t' generate G, hence bijective by finiteness. Its generator images make it unique.

For an inconsistent edge, let u be the recorded word reaching its initial vertex and let v be the recorded word reaching its final vertex. If the edge is labelled $z \in \{s, t\}$, the word uzv^{-1} evaluates to the identity in the source. The two conflicting target images show that it does not evaluate to the identity in the target. This gives the claimed obstruction, which can be checked by permutation multiplication without repeating the propagation. □

Proposition 2. *The supplied permutations give at least 46 distinct ordered chiral marking classes of rank three in* $\mathrm{PSU}_4(2)$*. Their ordered types and multiplicities are as follows.*

Type	Count	Type	Count	Type	Count	Type	Count
$(4, 9)$	2	$(5, 9)$	6	$(9, 4)$	2	$(12, 4)$	2
$(4, 12)$	2	$(5, 12)$	2	$(9, 5)$	6	$(12, 5)$	2
$(6, 9)$	2	$(9, 6)$	2	$(9, 12)$	4	$(12, 9)$	4
$(6, 12)$	2	$(9, 9)$	4	$(12, 6)$	2	$(12, 12)$	2

Proof. The certificate contains 92 literal tuples $(1, s, t)$ on forty points. The validator first closes the acquired ambient generators under permutation multiplication and obtains 25920 elements. For each tuple it verifies membership, the displayed order conditions, the three cyclic intersections

and equality of its generated subgroup with the ambient group. These are finite set equalities on actual permutations.

Apply Lemma 1 with target (s^{-1}, t^{-1}) to each tuple. Every propagation yields a relation obstruction. The retained 92 words therefore exclude every required reflection. The acquired recognition theorem now supplies a chiral polytope marking in each case.

To distinguish the markings, process them in their retained order. Compare a tuple with the existing class representatives of its same ordered type using Lemma 1; different types cannot be related by an automorphism. A successful propagation assigns the tuple to that representative. If all comparisons fail, retain a new representative. Induction shows that every new representative is inequivalent to all earlier ones, and every assigned tuple is equivalent to its recorded representative. The completed comparisons give 46 classes, each containing two of the supplied tuples. There are 46 successful transports, each checking 25920 vertices and 51840 directed edges, and 118 unsuccessful transports with explicit relation witnesses. The classes have precisely the sixteen distributions in the table. Neither a computed outer automorphism order nor completeness of the search is needed for this argument. □

For example, the first marking has type $(12, 9)$ and is given by the following permutations, with omitted points fixed:

$$\begin{aligned} s = {} & (1,24)(2,26,3,31)(5,12,10) \\ & (6,40,9,15,13,35,7,17,8,33,11,22) \\ & (14,38,37,16,19,36,32,18,21,34,39,20) \\ & (23,25)(27,28,29,30), \\ t = {} & (1,2)(3,4)(5,14)(6,20)(7,17)(8,15)(9,21) \\ & (10,18)(11,16)(12,22)(13,19)(23,32)(24,38) \\ & (25,35)(26,33)(27,39)(28,36)(29,34)(30,40)(31,37). \end{aligned}$$

One reflection obstruction is

$$w(s,t) = s^2ts^5tst^{-1}s^{-1}t^{-1}s^{-3}t^{-1} = 1, \qquad w(s^{-1}, t^{-1}) \neq 1.$$

Both statements are checked by direct multiplication with permutations acting on the right. The complete certificate supplies the same form of witness for every proposed reflection and every rejected comparison.

Source comparison and attribution. Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, print 26 in the chiral column of the PSU$(4, 2)$ row of Table 6, on page 381. Section 4 classifies the ordered generator tuples up to isomorphism. Section 6 explicitly counts the two enantiomorphic forms separately, and halves its overall total only when these are subsequently identified. Thus the printed column uses the same convention as the proposition: its value 26 is contradicted by the literal lower bound 46. This comparison does not identify duals. The mirror formula in Section 3 becomes (s^{-1}, t) in our prefix coordinates, exactly the reflection tested above. The broader search reports the same count 46, but its exhaustion is not used in this refutation of the printed value.

The geometric reconstruction and named-group identification are borrowed, with the attributions retained in the acquired manuscripts. The present contribution is the explicit finite collection and its verification by permutation closure and relation words. This verifies a positive lower bound in ordinary Python, not a Lean formalization or a new general reconstruction theorem. No compression factor or completion of the whole source follows from this component alone.